

第二讲：相机模型与图像



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知识回顾

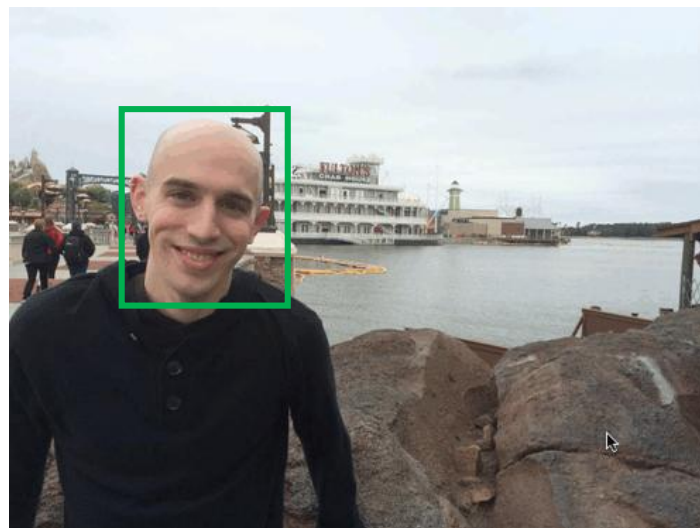
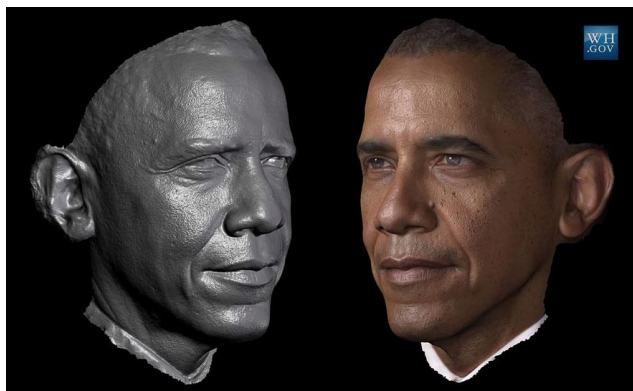


- 机器人视觉的用途 (3个R)

重建(Reconstruction): 从视觉信息中重建三维世界的特性

认知(Recognition): 使机器人识别环境中的物体、人、场景和活动等

重组(Reorganization): 感知构造, 如分组与分割等



图像的产生

- 图形与图像的区别
 - 图形是一个物体的三维模型
 - 图像是由三维模型生成的一幅二维图像



视觉技术



成像方式或
图形学渲染



图像

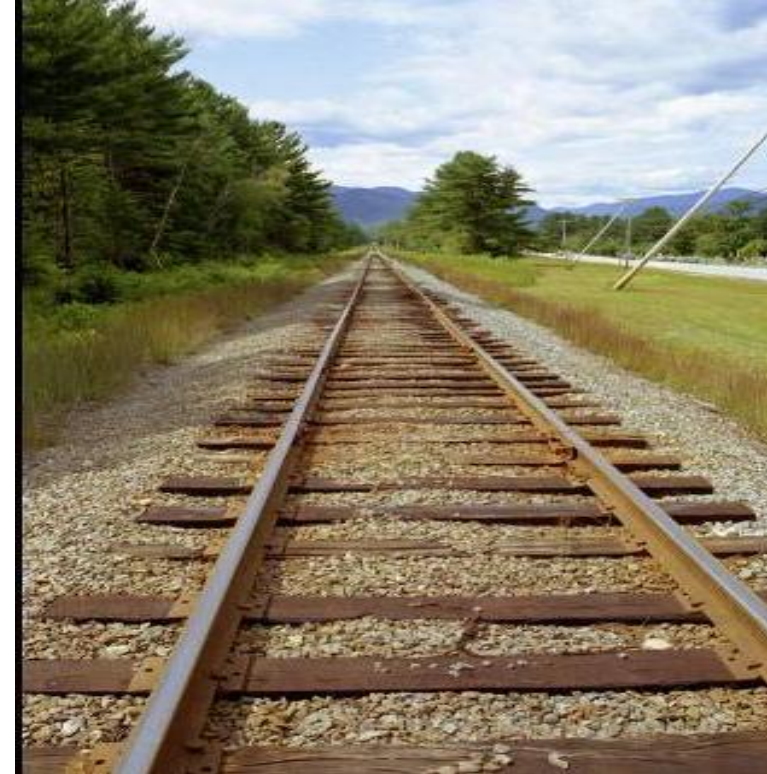
目录



- 1, 相机模型
 - 相机内部参数、外部参数
 - 相机的组成
- 2, 图像与颜色
- 3, 图像处理基础

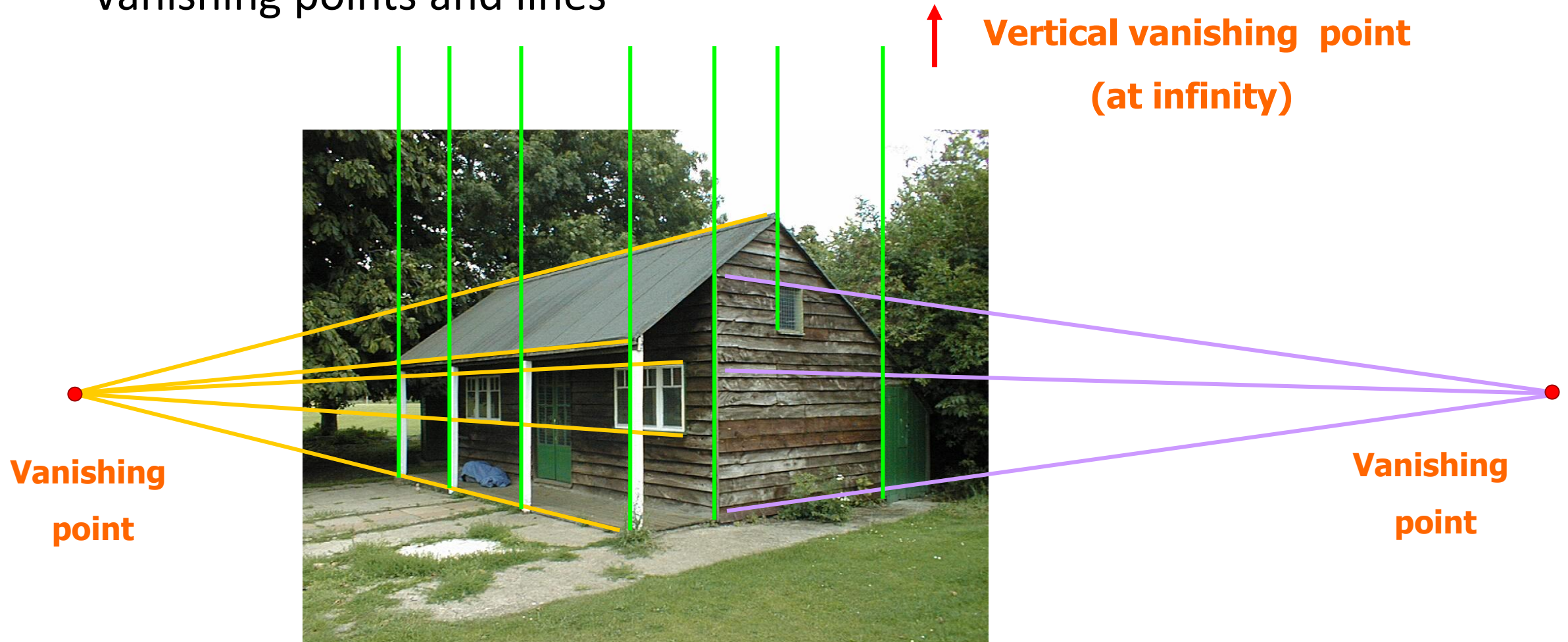
Perspective / 透视

- Parallel lines in the world intersect in the projected image at a “vanishing point”

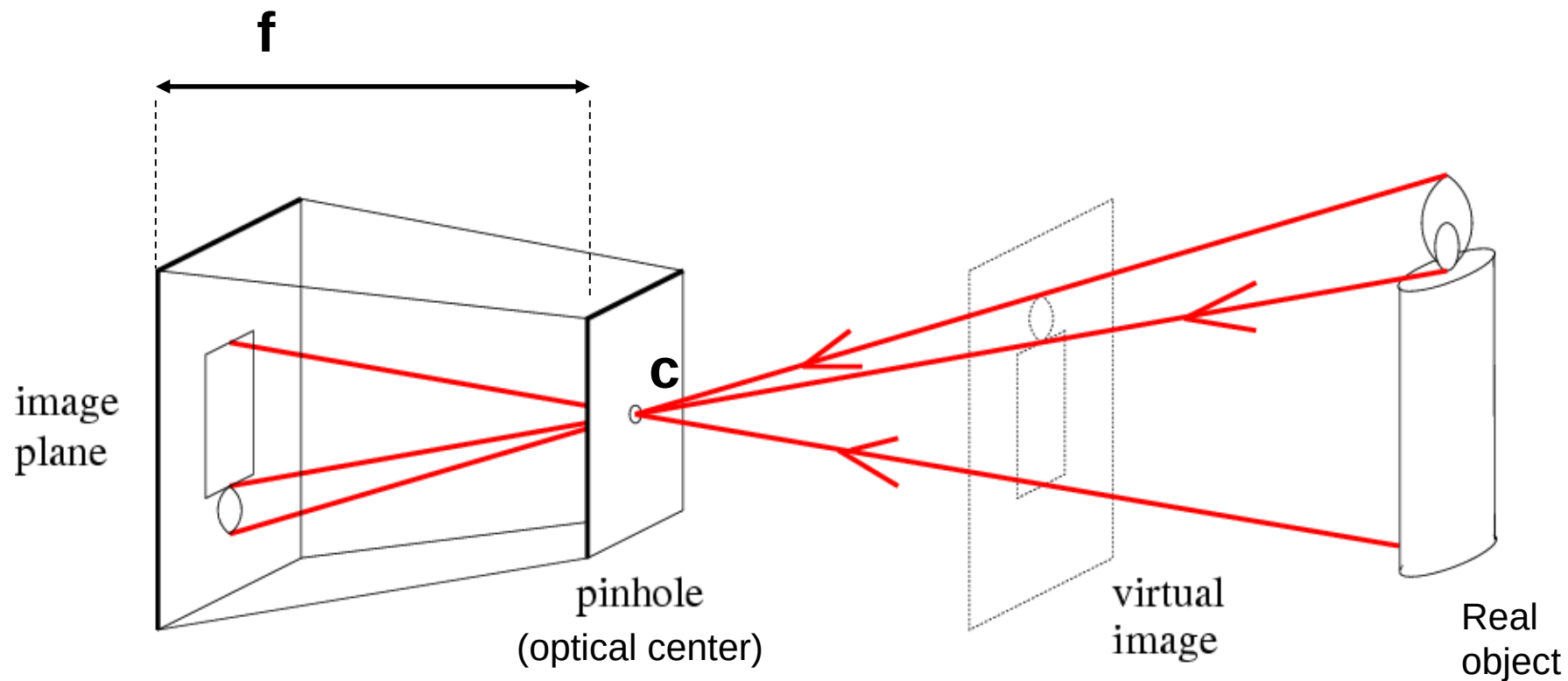


Perspective 透视

- Vanishing points and lines



针孔相机模型(Pinhole camera model)

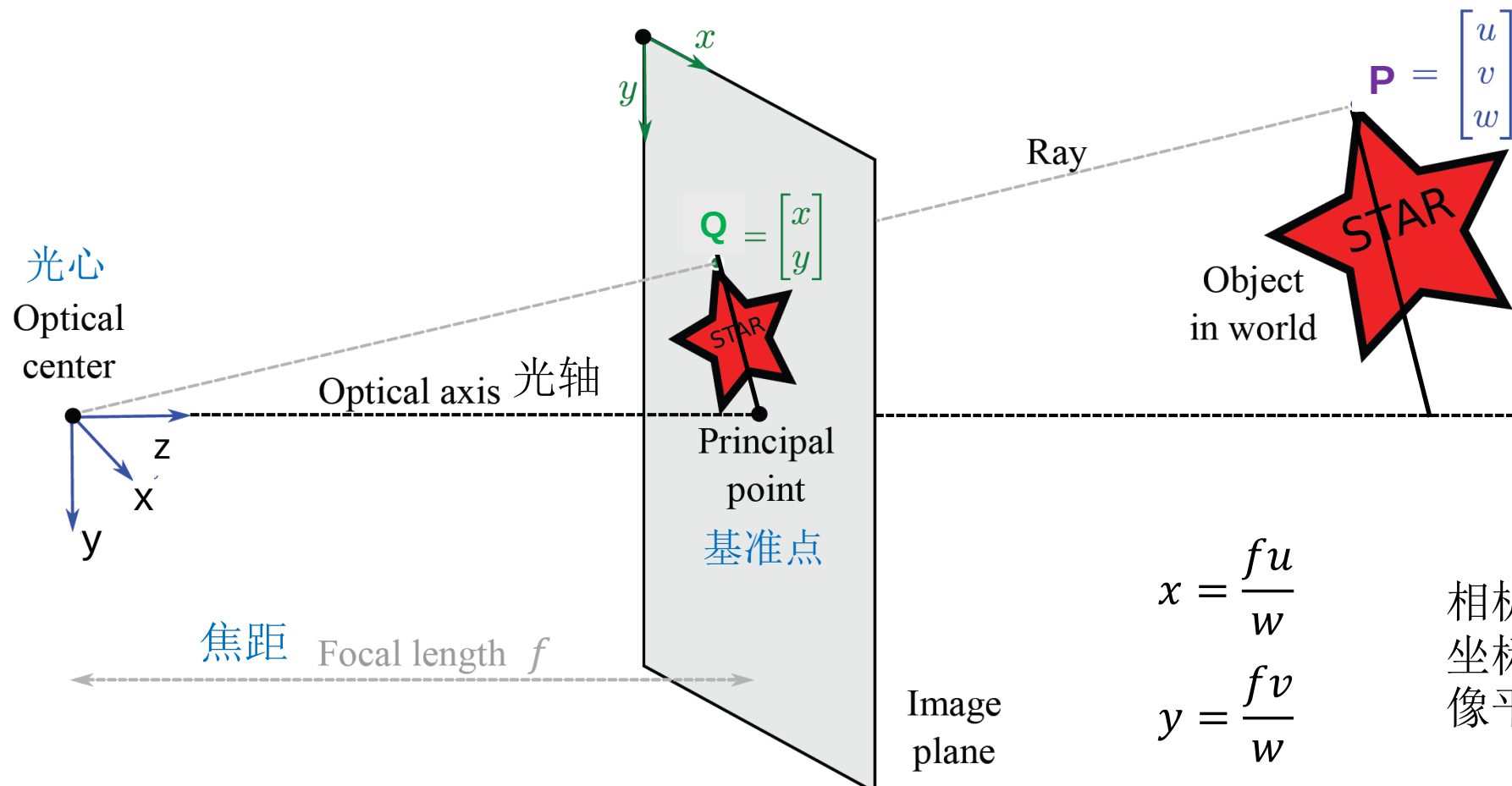


焦距 f = Focal length

光心 c = Optical center of the camera

针孔相机模型(Pinhole camera model)

- 透视投影



相机坐标系下，**光心**是坐标原点，**基准点**是成像平面内的坐标原点。

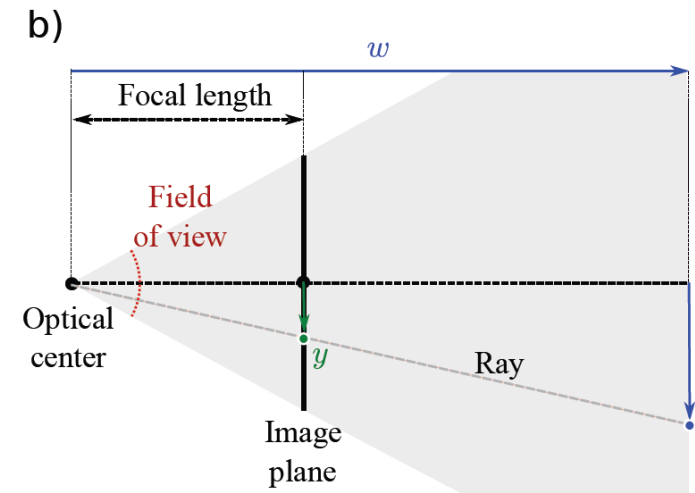
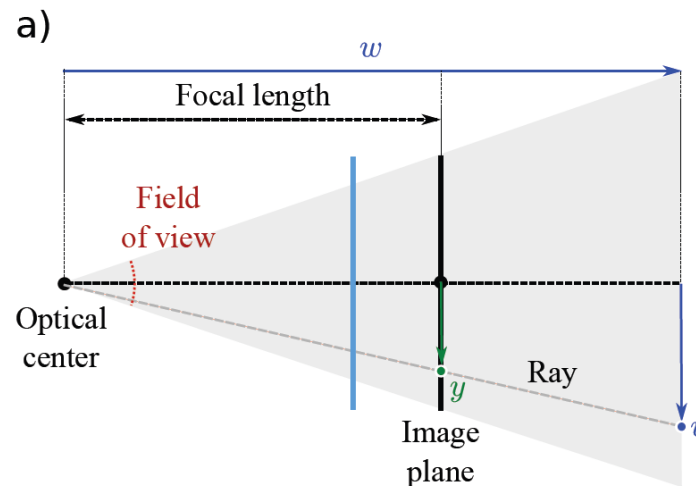
针孔相机模型(Pinhole camera model)

影响投影平面内坐标的因素

- 焦距
- 光感受器的密度

$$x = \frac{fu}{w} \quad y = \frac{fv}{w}$$

这两者对x和y的值的影响都是线性的，因此可以用一个单一的参数 ϕ 来表示二者的组合结果。



针孔相机模型(Pinhole camera model)

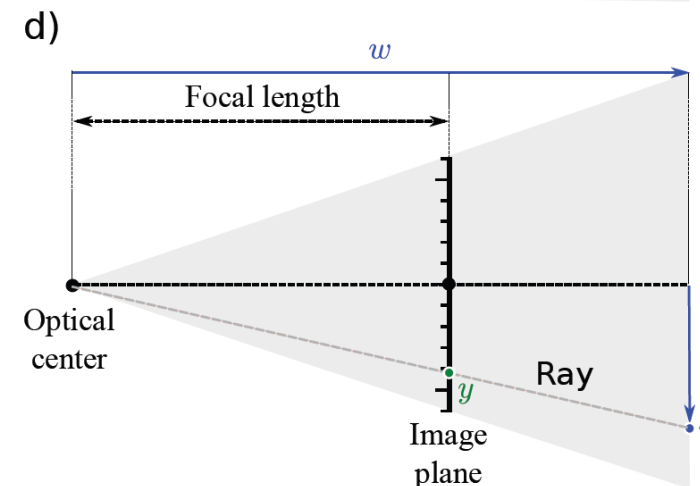
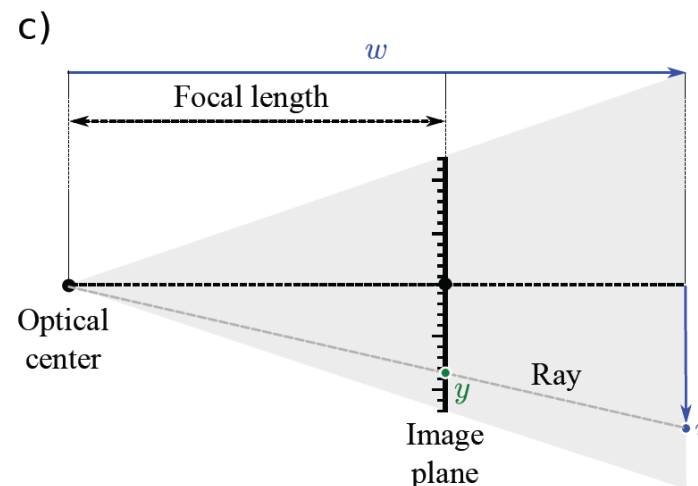
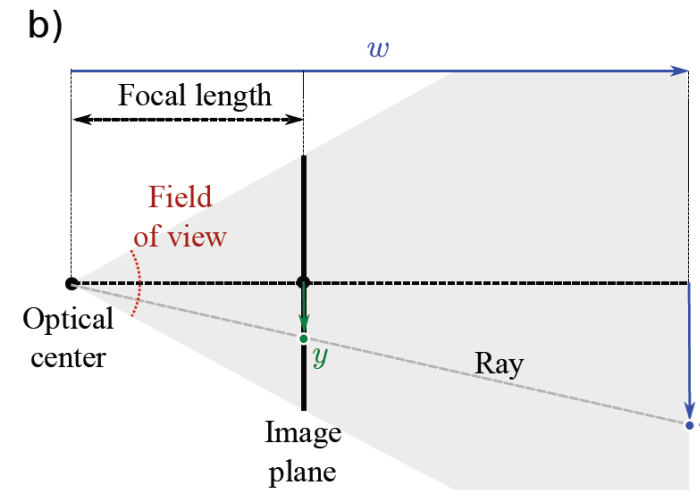
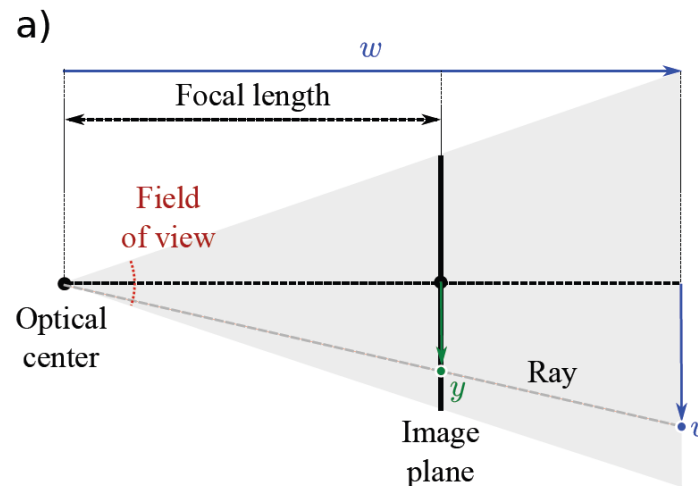


影响投影平面内坐标的因素

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针孔相机模型(Pinhole camera model)

用一个单一的焦距参数 ϕ 对以下二者进行表示

- 成像平面到光心的距离 f
- 光感受器的密度

$$x = \frac{fu}{w}$$

$$y = \frac{fv}{w}$$



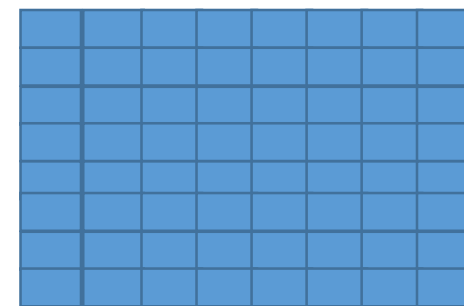
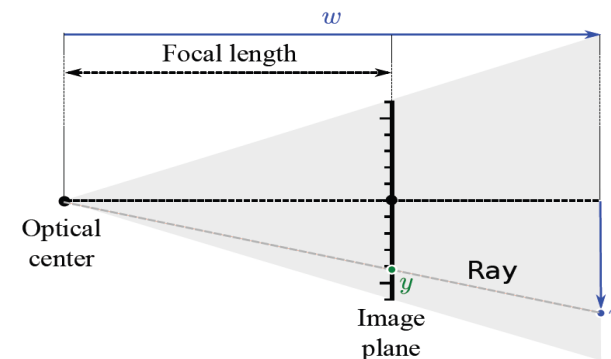
$$x = \frac{\phi u}{w}$$

$$y = \frac{\phi v}{w}$$

实际上, 光感受器的密度在不同方向可能有差异, 因此在x和y方向上用不同的焦距参数

$$x = \frac{\phi_x u}{w}$$

$$y = \frac{\phi_y v}{w}$$



像素网格

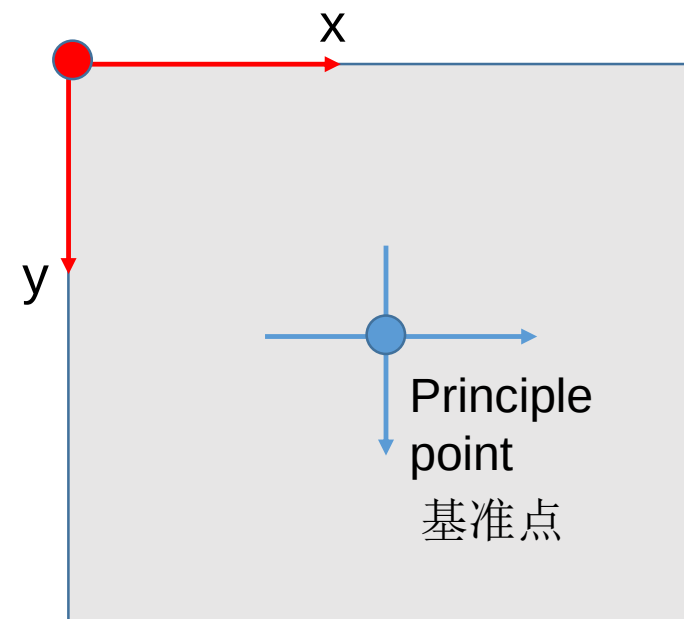
针孔相机模型(Pinhole camera model)



偏移参数

- 上述模型假设图像平面的**原点(0,0)** 位于成像平面的**中心**（即基准点——光轴与成像平面的交点）
- 若将图像平面的原点定义在**左上角**，则有一个偏移量：

$$\begin{aligned}x &= \frac{\phi_x u}{w} + \delta_x \\y &= \frac{\phi_y v}{w} + \delta_y\end{aligned}$$



针孔相机模型(Pinhole camera model)

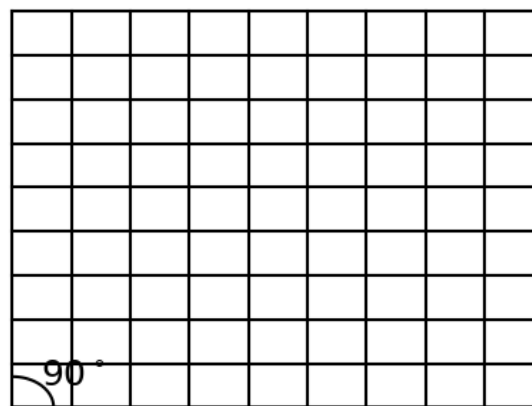


- 斜切参数 (Skew parameter)
 - 考虑投影平面的坐标轴不垂直的情况

$$x = \frac{\phi_x u + \gamma v}{w} + \delta_x$$
$$y = \frac{\phi_y v}{w} + \delta_y$$

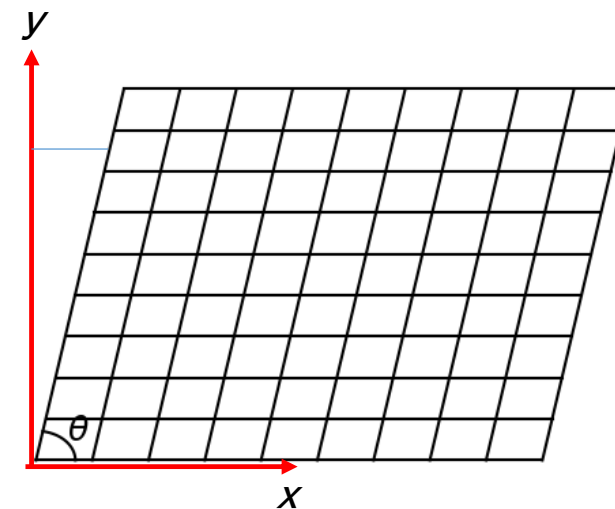
相机内部参数
(Intrinsic parameters)

$\phi_x, \phi_y, \delta_x, \delta_y, \gamma$



Ideally aligned pixel grid

理想的对齐的像素网格

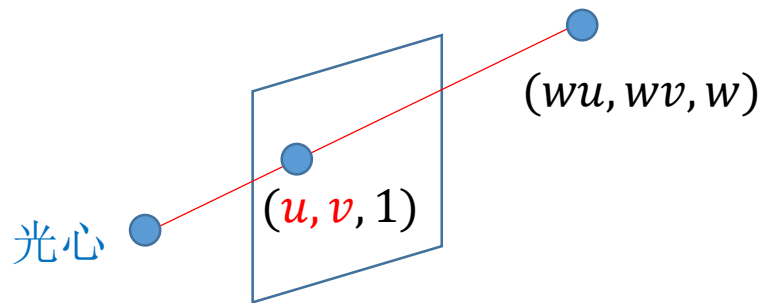


Skewed pixel grid

歪斜的像素网格

齐次坐标(Homogeneous coordinates)

- 齐次坐标方便用矩阵乘法表示坐标之间的转换关系
 - 成像平面的一个点，对应三维空间的一条射线



常规坐标转为齐次坐标

$$\begin{matrix} \text{in } \mathbb{R}^2 \\ (u, v) \end{matrix} \Rightarrow \begin{matrix} \text{in } \mathbb{P}^2 \\ \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \end{matrix} \sim \begin{matrix} \text{in } \mathbb{P}^2 \\ \begin{bmatrix} wu \\ wv \\ w \end{bmatrix} \end{matrix}$$

$$\begin{matrix} \text{in } \mathbb{R}^3 \\ (X, Y, Z) \end{matrix} \Rightarrow \begin{matrix} \text{in } \mathbb{P}^3 \\ \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \end{matrix} \sim \begin{matrix} \text{in } \mathbb{P}^3 \\ \begin{bmatrix} wX \\ wY \\ wZ \\ w \end{bmatrix} \end{matrix}$$

- 齐次坐标转为常规坐标

$$\begin{matrix} \text{in } \mathbb{P}^2 \\ \begin{bmatrix} x \\ y \\ w \end{bmatrix} \end{matrix} \Rightarrow \begin{matrix} \text{in } \mathbb{R}^2 \\ \left(\frac{x}{w}, \frac{y}{w} \right) \end{matrix}$$

$$\begin{matrix} \text{in } \mathbb{P}^3 \\ \begin{bmatrix} X \\ Y \\ Z \\ w \end{bmatrix} \end{matrix} \Rightarrow \begin{matrix} \text{in } \mathbb{R}^3 \\ \left(\frac{X}{w}, \frac{Y}{w}, \frac{Z}{w} \right) \end{matrix}$$

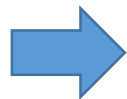
针孔相机模型(Pinhole camera model)



相机内部参数

原始坐标

$$\begin{aligned}x &= \frac{\phi_x u + \gamma v}{w} + \delta_x \\y &= \frac{\phi_y v}{w} + \delta_y\end{aligned}$$



齐次坐标

$$\begin{aligned}x &= \phi_x u + \gamma v + \delta_x w \\y &= \phi_y v + \delta_y w \\z &= w\end{aligned}$$



齐次坐标的矩阵形式

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \phi_x & \gamma & \delta_x \\ 0 & \phi_y & \delta_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

相机的内参可以表示为一个3x3的矩阵 K
 K 为一个上三角矩阵

$$x = Ku$$

图像中的坐标

三维空间中的坐标

针孔相机模型(Pinhole camera model)

相机的外部参数：位置和方向

- 目前，我们考虑的是相机坐标系与世界坐标系重合的情况
- 相机本身会在世界坐标系中移动和旋转

相机坐标系
下的坐标

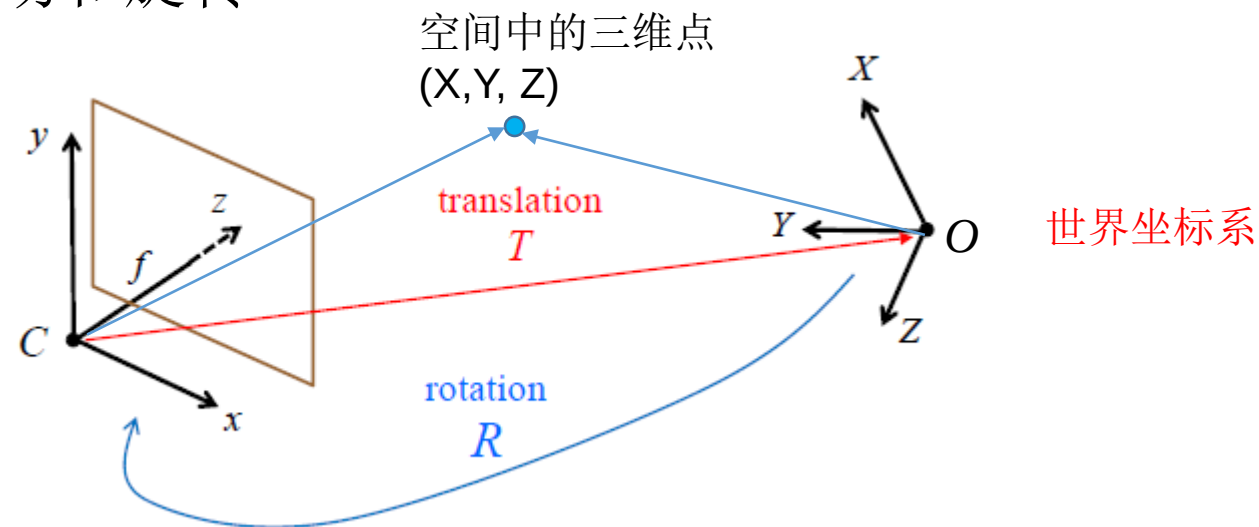
世界坐标系
下的坐标

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = R \cdot \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + T$$



$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1} = \underbrace{\begin{bmatrix} R & T \end{bmatrix}}_{3 \times 4} \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}_{4 \times 1}$$

相机坐标系



通过齐次坐标下的
线性变换表示

针孔相机模型(Pinhole camera model)



- 内部 (intrinsic) 参数

$$\{\phi_x, \phi_y, \delta_x, \delta_y, \gamma\}$$

$$K = \begin{bmatrix} \phi_x & \gamma & \delta_x \\ 0 & \phi_y & \delta_y \\ 0 & 0 & 1 \end{bmatrix}$$

- 外部 (Extrinsic) 参数

$$\{R, T\}$$

图像中的
二维点

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = K \cdot \underbrace{[R \mid T]}_{\text{Projection Matrix P}} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

世界坐标系
中的三维点

投影矩阵

K: project (5 d.o.f.)

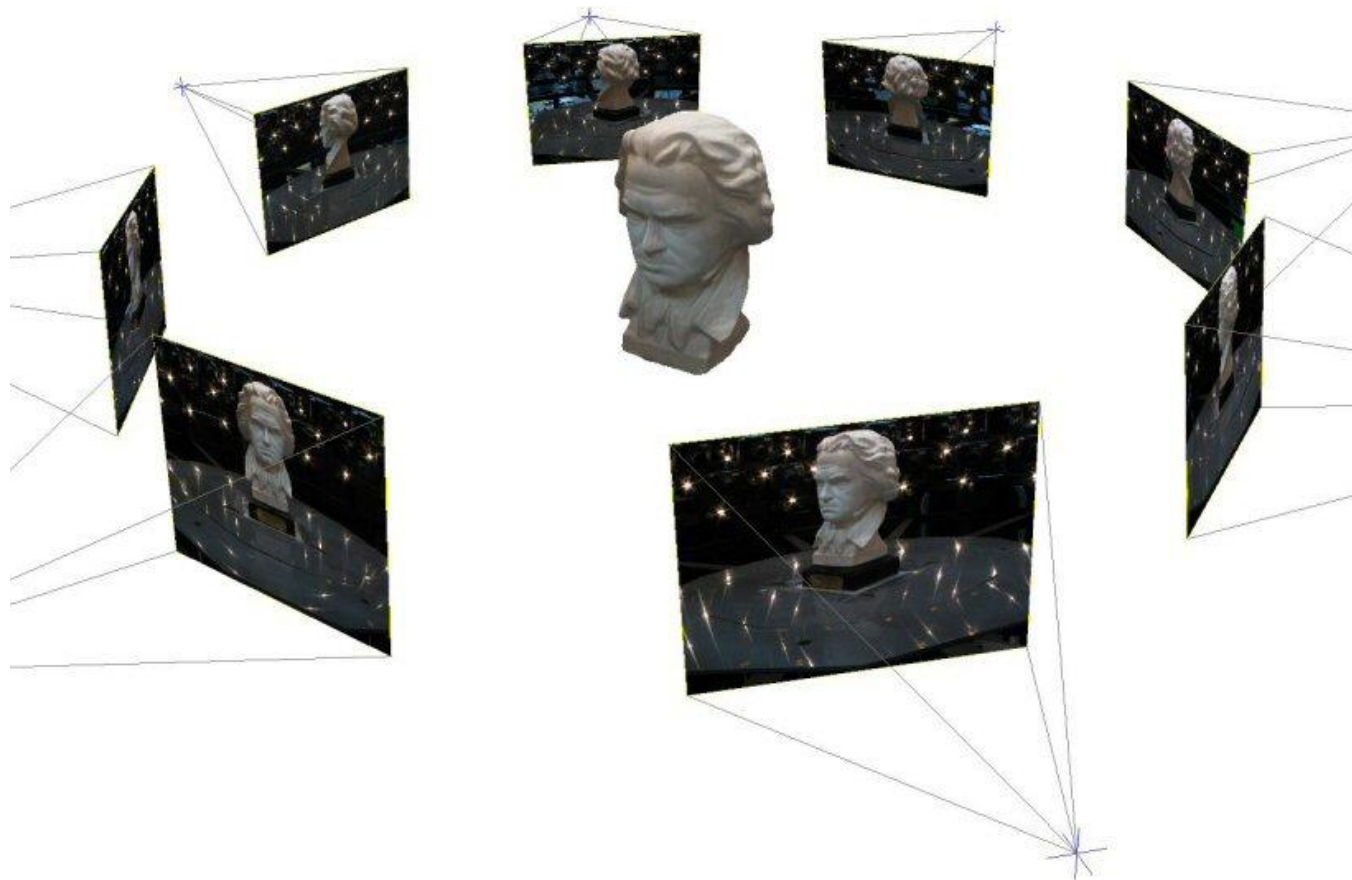
R: rotate (3 d.o.f.)

T: translate (3 d.o.f.)

相机模型的应用

多视角三维重建

- 从不同角度的照片中重建出三维模型
- 需要知道相机的内部参数和外部参数



More complex model

- Home-made pinhole camera

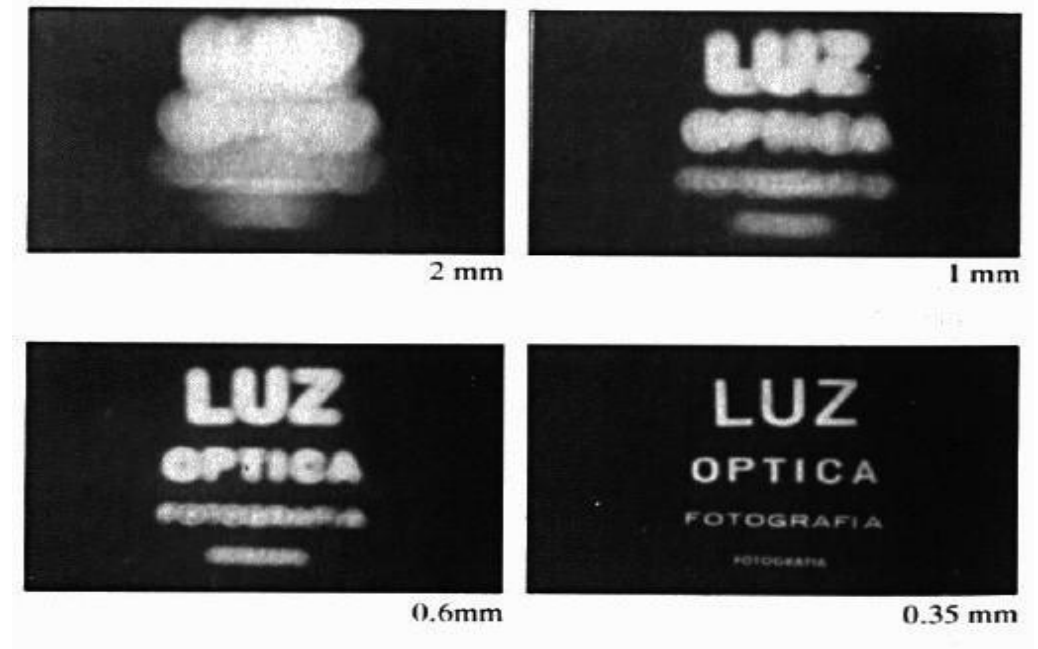
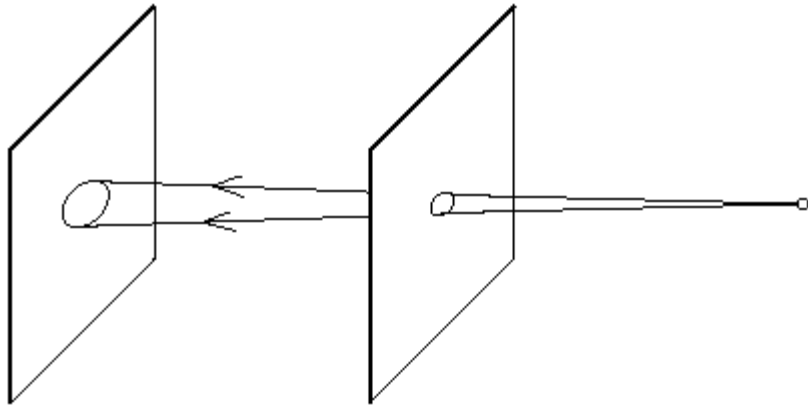


<http://www.debevec.org/Pinhole/>

More complex model

- Shrinking the aperture (光圈)

Integrate over fewer angles



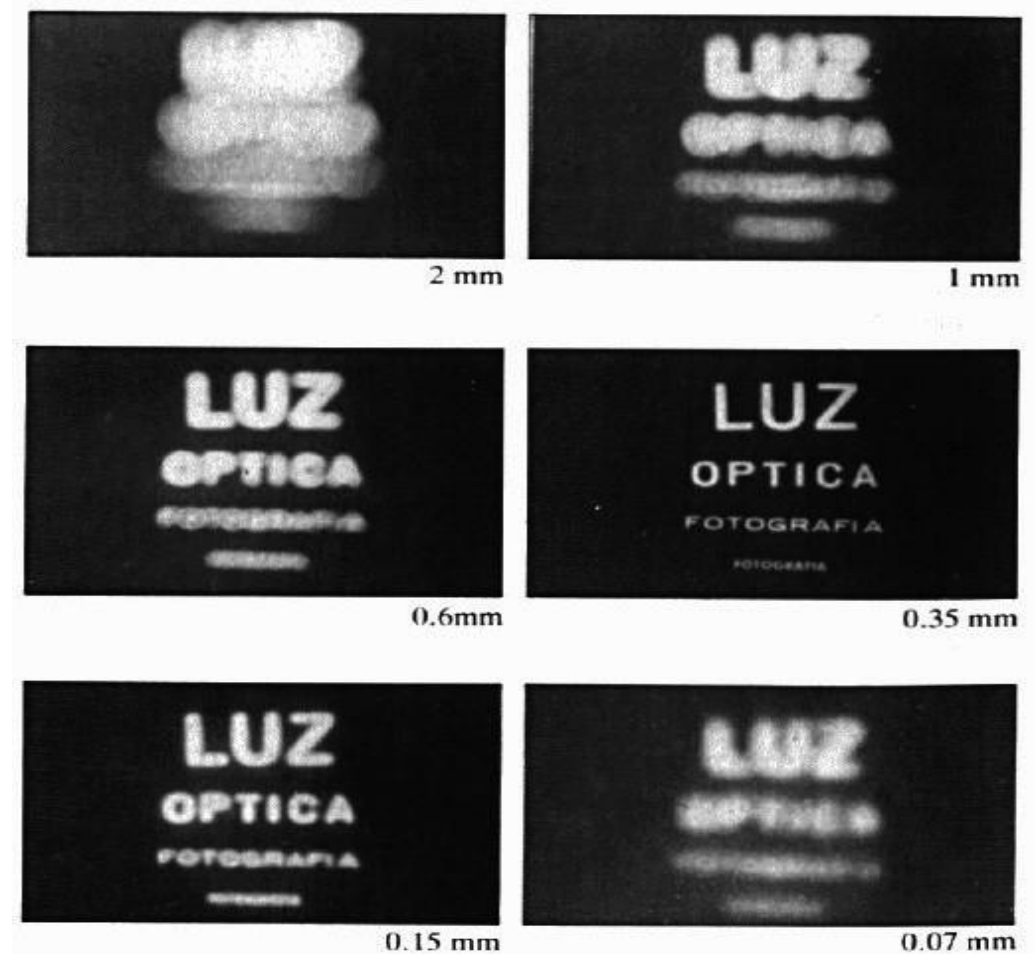
Less light gets through

More complex model

- Shrinking the aperture (光圈)

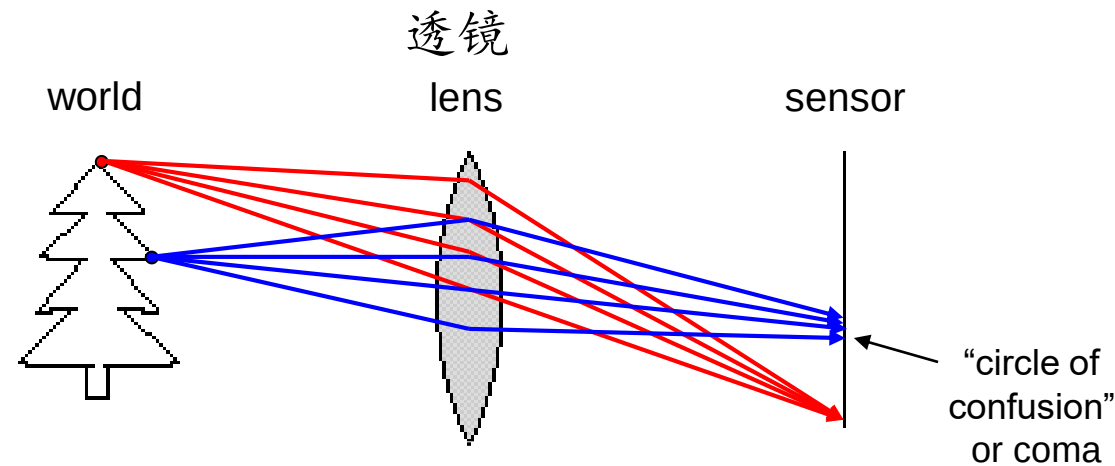
Why not make the aperture as small as possible?

- Less light gets through
- Diffraction effects... 衍射现象



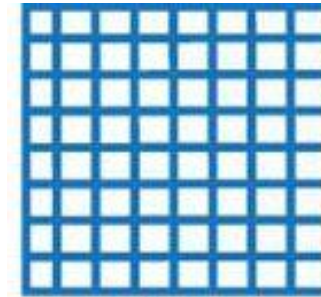
More complex model

- Focus and Defocus 聚焦和散焦
- A lens focuses light onto the film
 - There is a specific distance at which objects are “in focus”
 - other points project to a “circle of confusion” in the image
 - Changing the shape of the lens changes this distance



More complex model

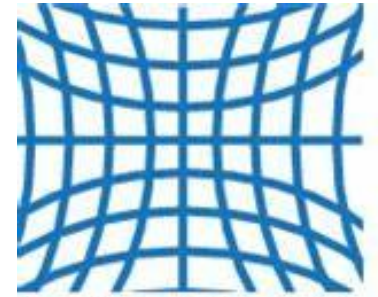
- Radical distortion 径向失真
 - Wide-angle lens 广角镜头



No distortion



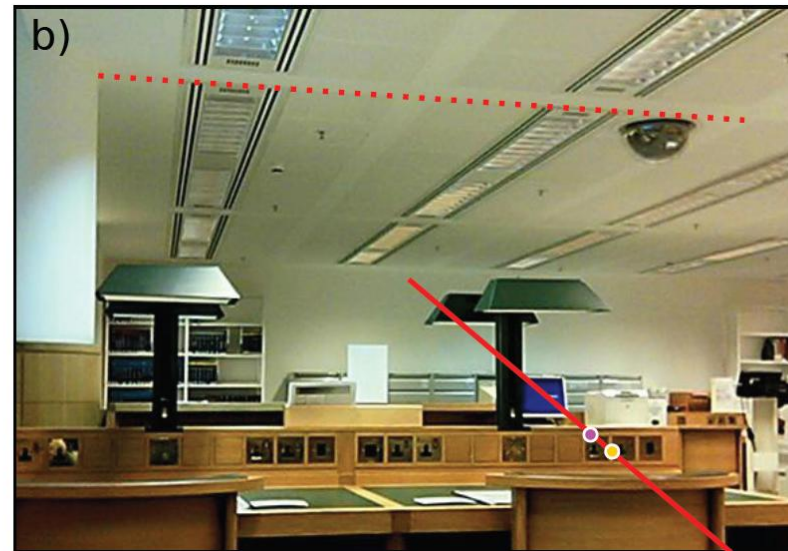
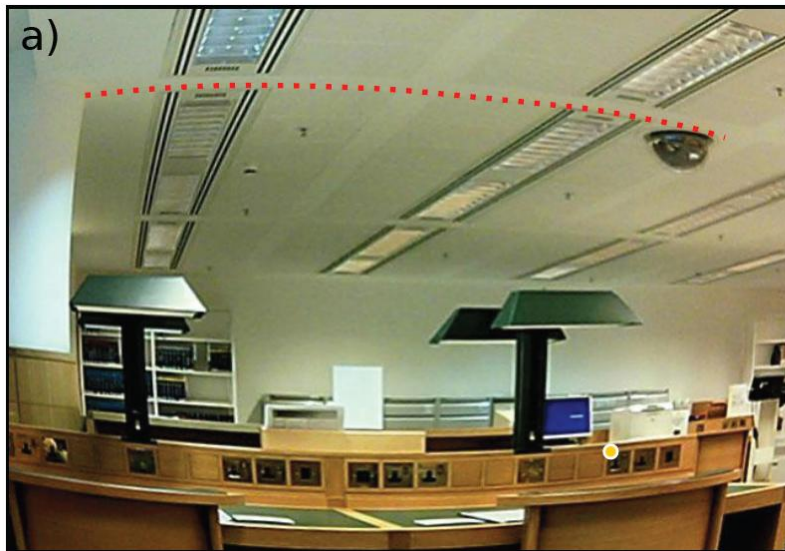
Positive radial distortion
"barrel"



Negative radial distortion
"pincushion"

桶形失真

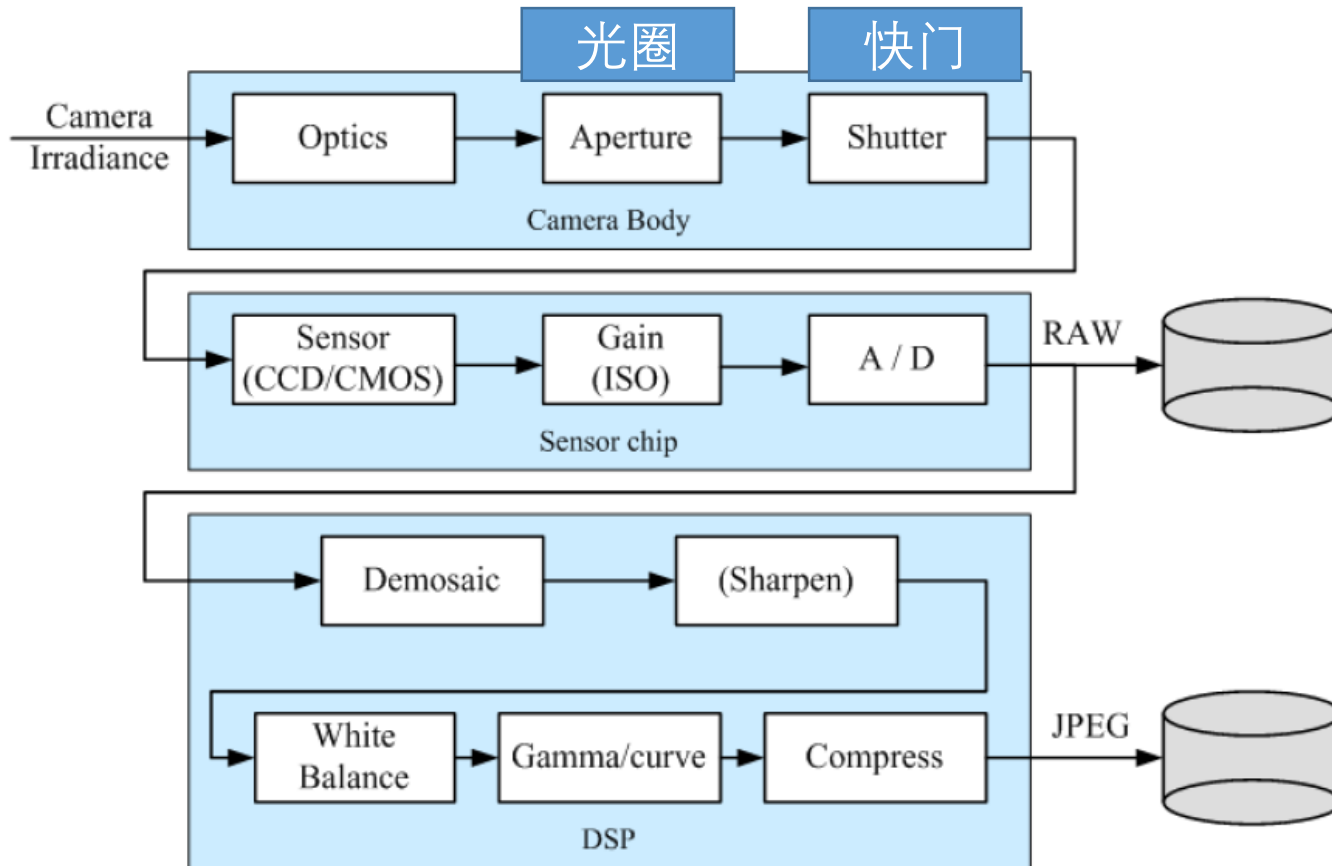
枕形失真



$$x' = x(1 + \beta_1 r^2 + \beta_2 r^4)$$

$$y' = y(1 + \beta_1 r^2 + \beta_2 r^4)$$

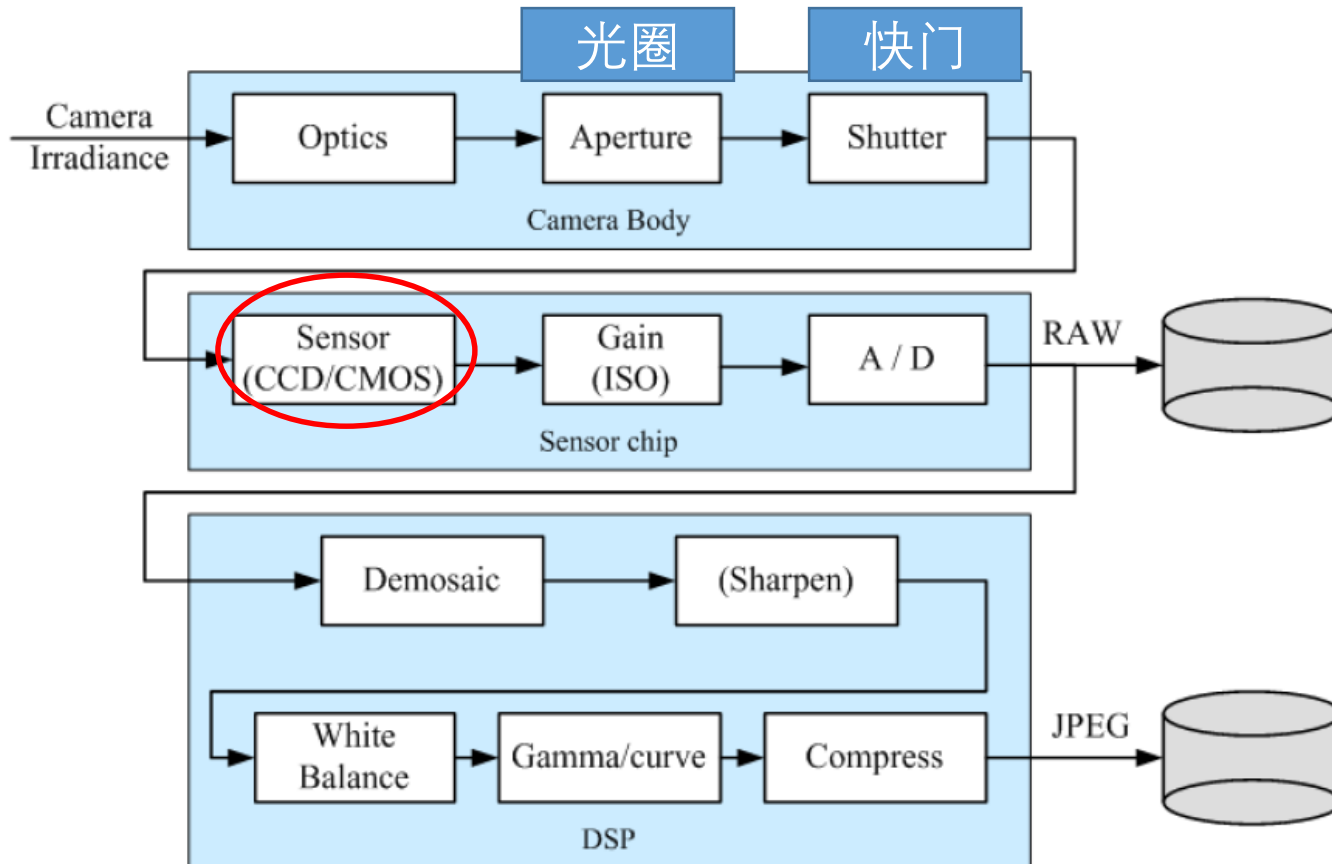
Digital Camera



Shutter 快门



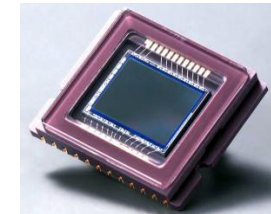
Digital Camera



传感器

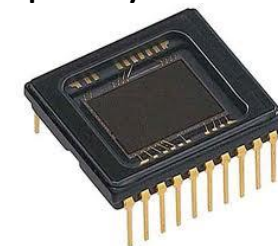
- CCD (charged couple device)
 - High quality but expensive

电荷耦合
器件

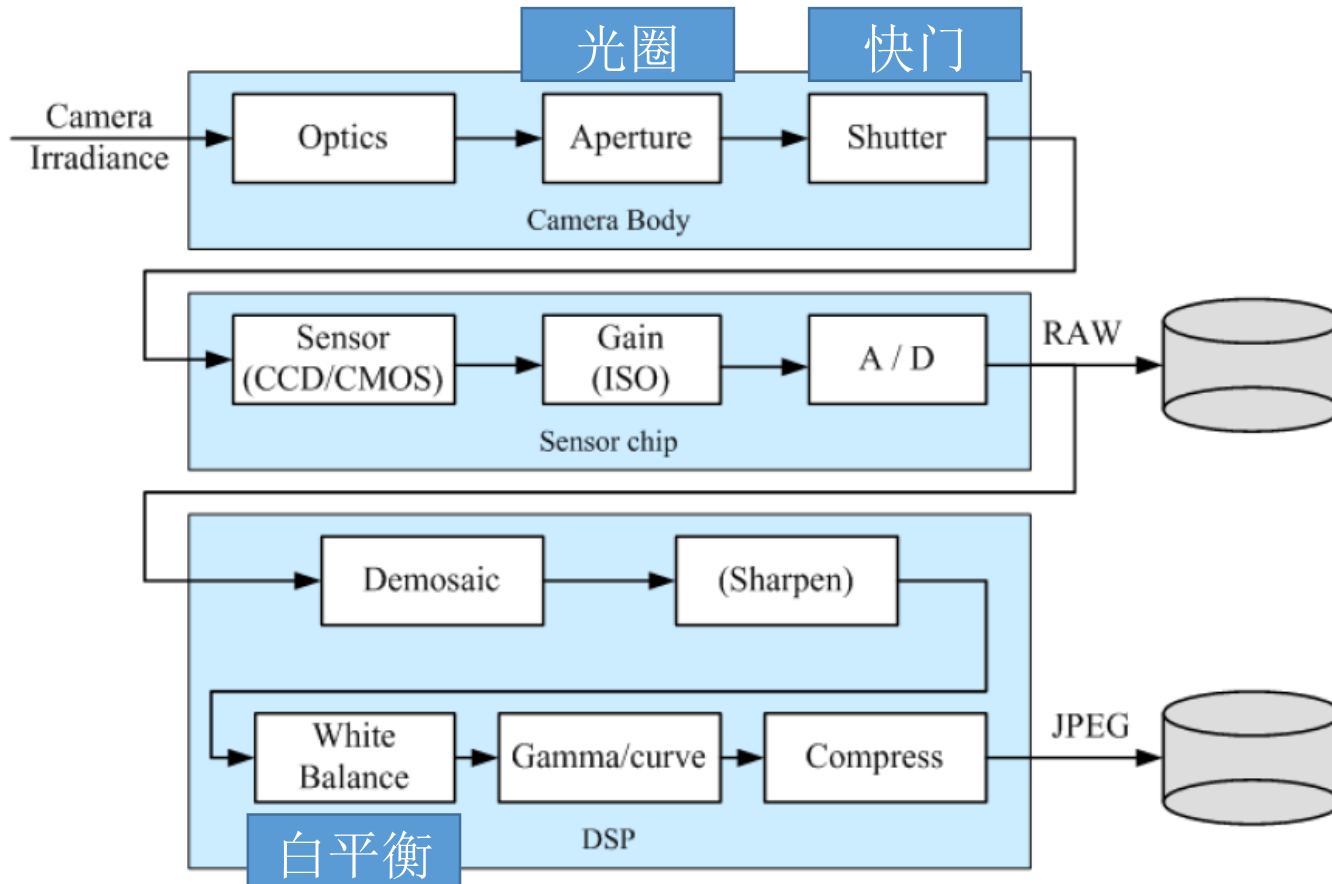


- CMOS (complimentary mental oxide semiconductor)
 - Lower quality but Cheaper

互补金属
氧化物半
导体



Digital Camera



- White balance 白平衡
 - Removing unrealistic color
 - Take into account the "color temperature"



目录

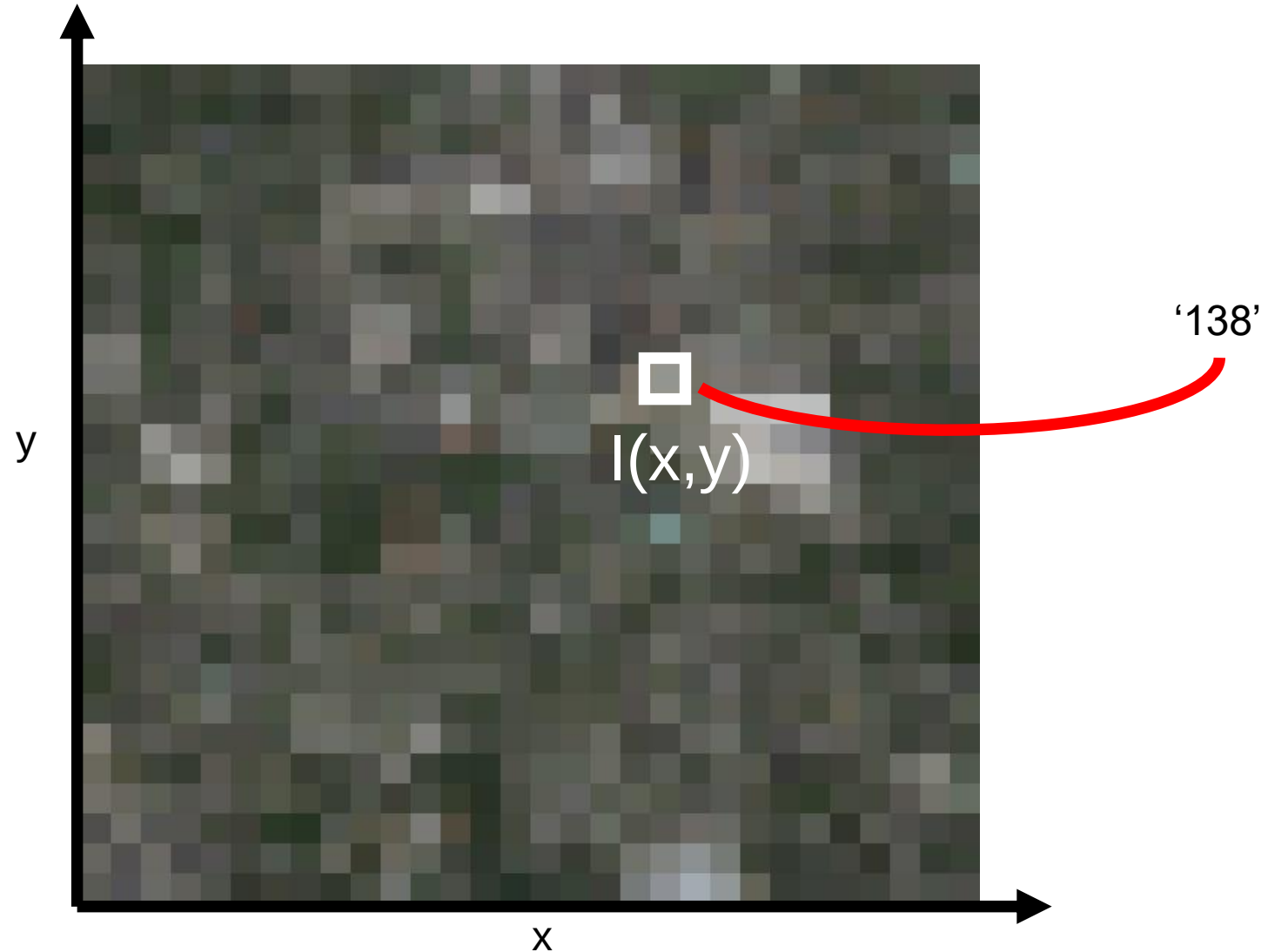


- 1, 相机模型
 - 相机内部参数、外部参数
 - 相机的组成
- 2, 图像与颜色
- 3, 图像处理基础

What is an image

- Image is stored as an array in computer
- Each element is a pixel
- Each pixel has an intensity value

图像在计算机中表示为矩阵的形式，矩阵中的每一个元素对应一个像素。



What is an image

Image as a 2D sampling of signal 图像是信号的一个二维采样

- Signal: function depending on some variable with physical meaning
- Image: sampling of the signal function
 - 2 variables: xy coordinates
 - 3 variables: xy + time (video)
 - 'Brightness' is the value of the function for visible light

对哪些信号的采样可以产生图像？

- Can be different physical values
 - Light, temperature, pressure, depth ...

Example 2D Images

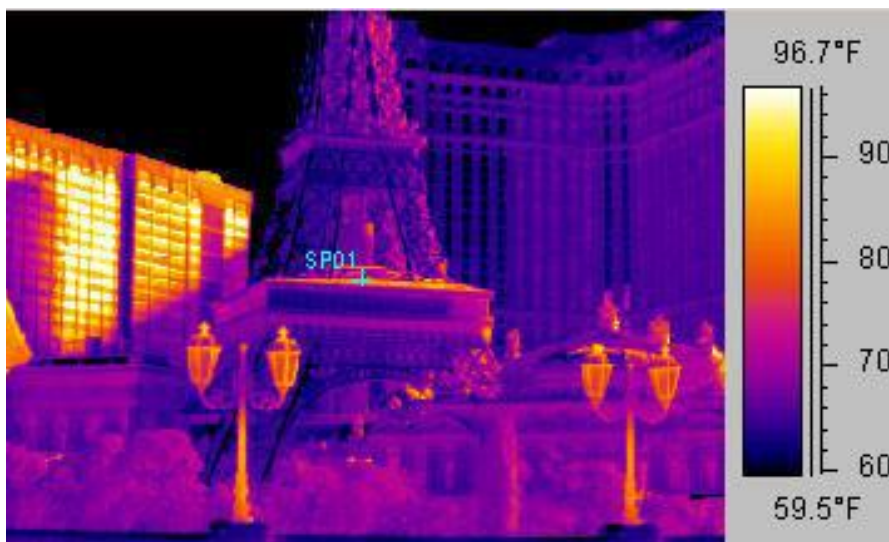
光学图像
(可见光)



超声图像
(声速/密度)



红外图像
(温度)



磁共振图像
(水分子含量)

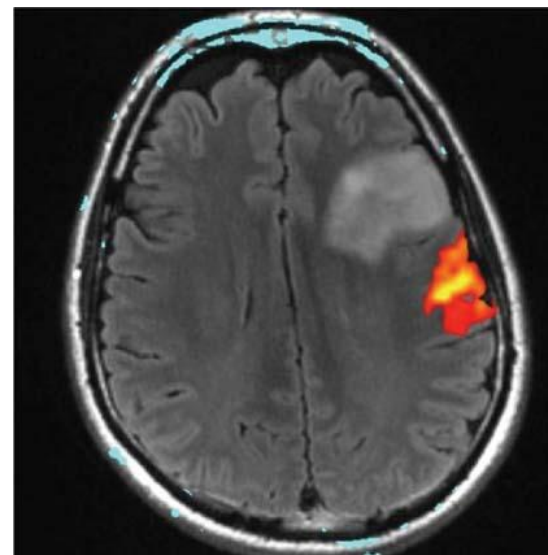


Image and sampling

- Sampling in 1D takes a function, and returns a vector whose elements are values of that function at the sample points.

对一维信号进行采样，得到一个向量

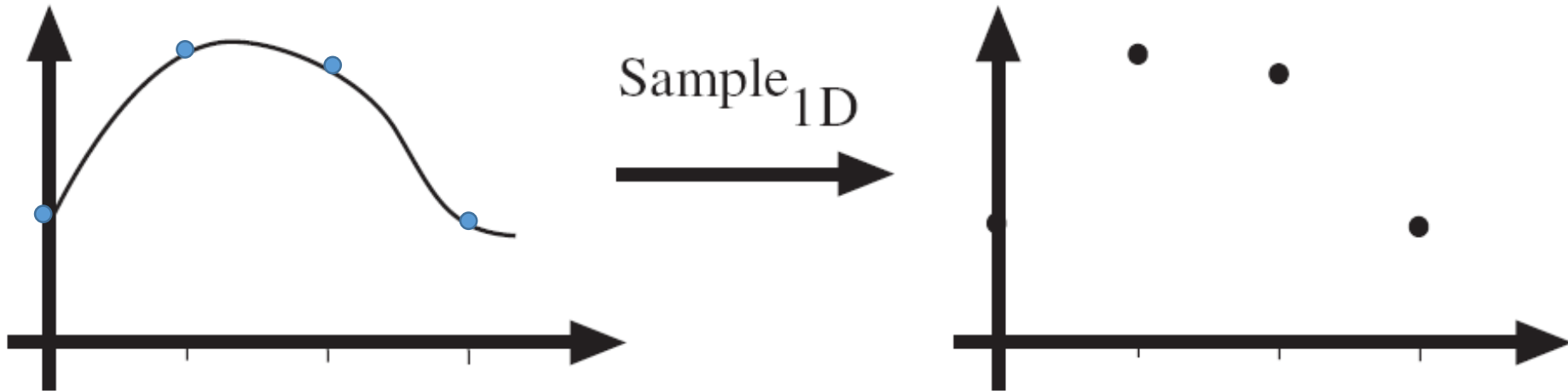
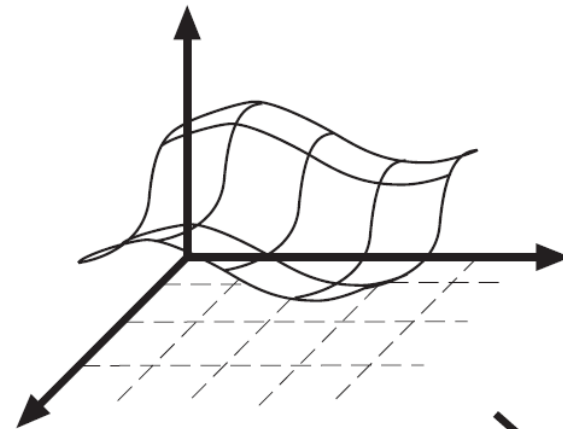


Image and sampling

Sampling in 2D takes a function and returns a matrix.



对二维信号进行采样，得到一个数组

Sample_{2D}

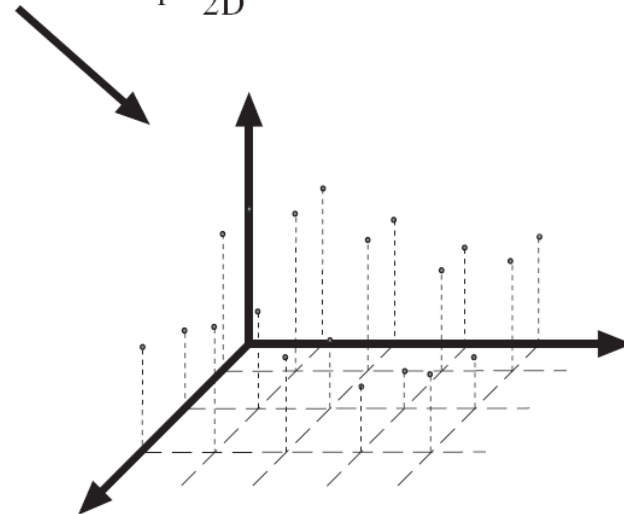


Image and sampling

Image is a result of sampling by camera sensor

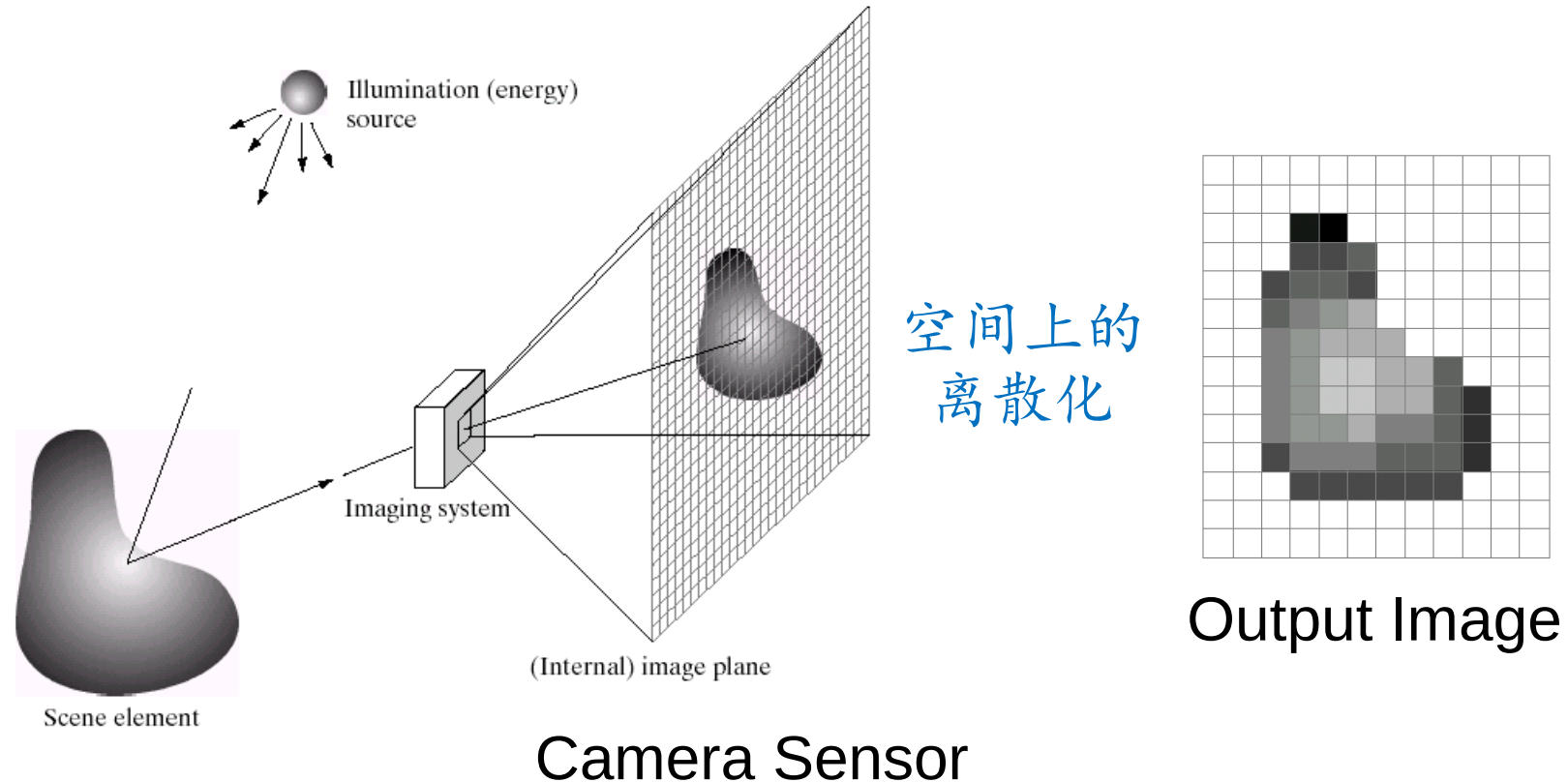
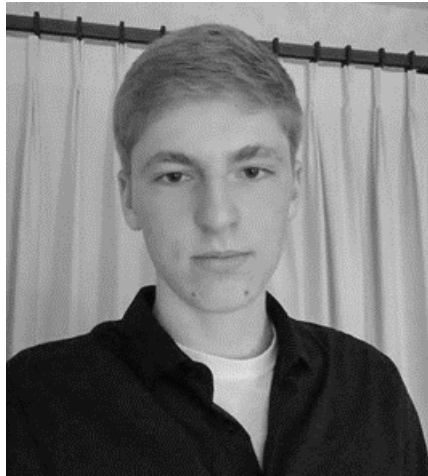


Image and sampling

Quantization Effects – Radiometric Resolution

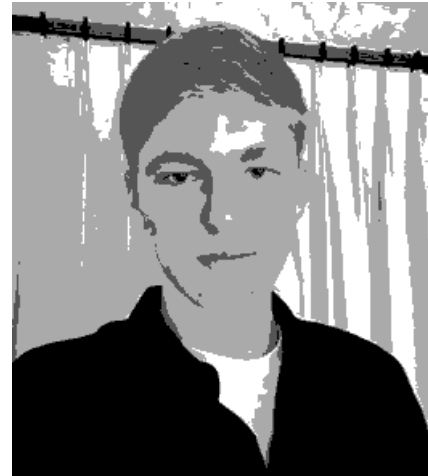
对像素值的
离散化



8 bit – 256 levels



4 bit – 16 levels



2 bit – 4 levels



1 bit – 2 levels

Image as array



Images in Python Numpy

N x M grayscale image “im”

- `im[0,0]` = top-left pixel value
- `im[y, x]` = y pixels down, x pixels to right
- `im[N-1, M-1]` = bottom-right pixel

it

Column

M

Row

N

0.92	0.93	0.94	0.97	0.62	0.37	0.85	0.97	0.93	0.92	0.99
0.95	0.89	0.82	0.89	0.56	0.31	0.75	0.92	0.81	0.95	0.91
0.89	0.72	0.51	0.55	0.51	0.42	0.57	0.41	0.49	0.91	0.92
0.96	0.95	0.88	0.94	0.56	0.46	0.91	0.87	0.90	0.97	0.95
0.71	0.81	0.81	0.87	0.57	0.37	0.80	0.88	0.89	0.79	0.85
0.49	0.62	0.60	0.58	0.50	0.60	0.58	0.50	0.61	0.45	0.33
0.86	0.84	0.74	0.58	0.51	0.39	0.73	0.92	0.91	0.49	0.74
0.96	0.67	0.54	0.85	0.48	0.37	0.88	0.90	0.94	0.82	0.93
0.69	0.49	0.56	0.66	0.43	0.42	0.77	0.73	0.71	0.90	0.99
0.79	0.73	0.90	0.67	0.33	0.61	0.69	0.79	0.73	0.93	0.97
0.91	0.94	0.89	0.49	0.41	0.78	0.78	0.77	0.89	0.99	0.93

Image as array

Grayscale intensity

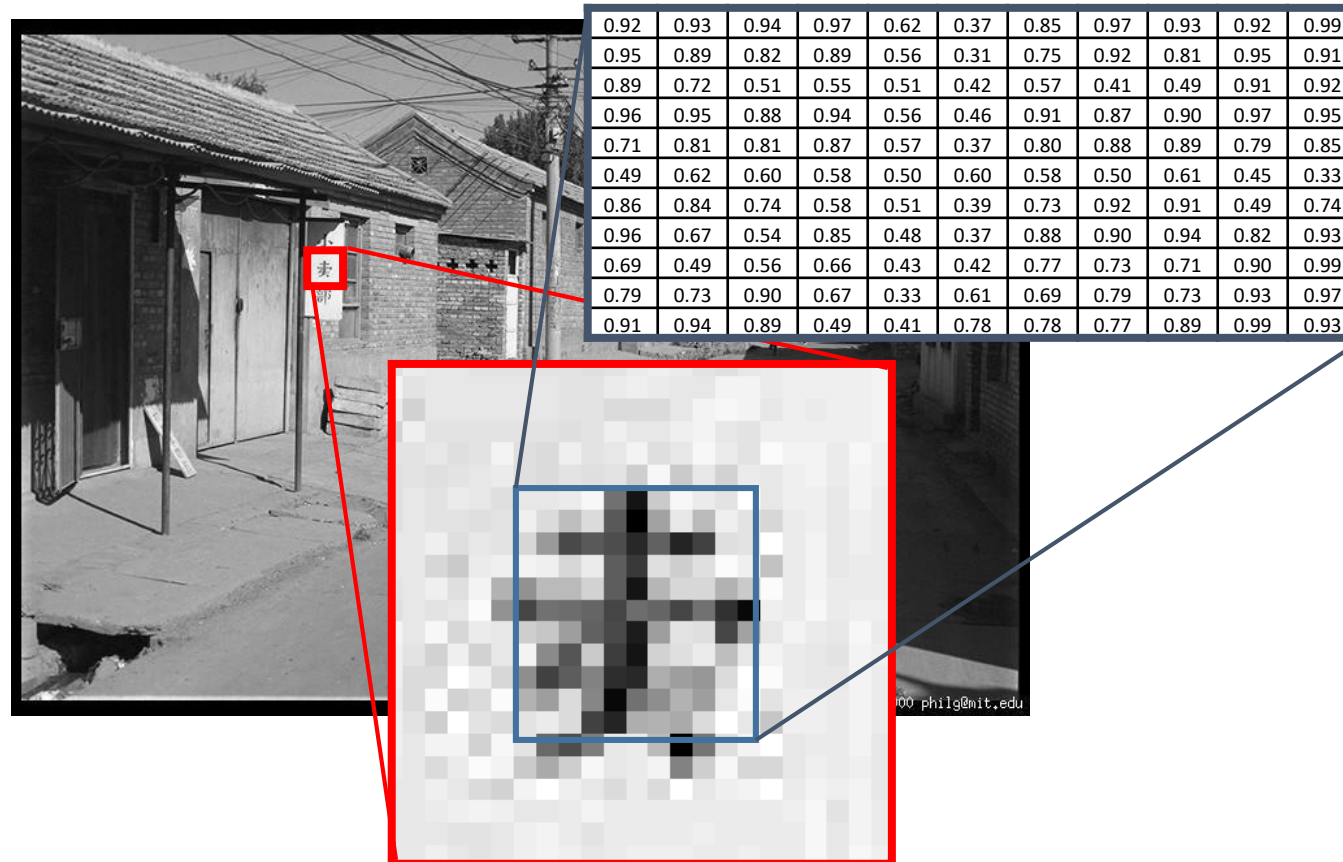
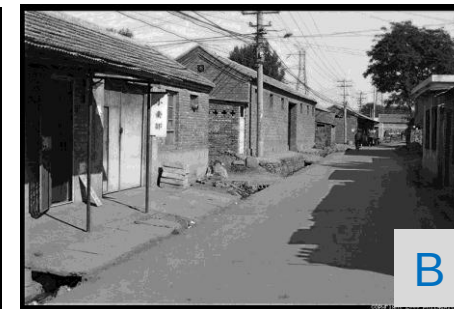
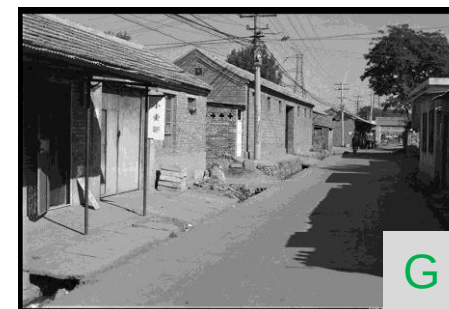
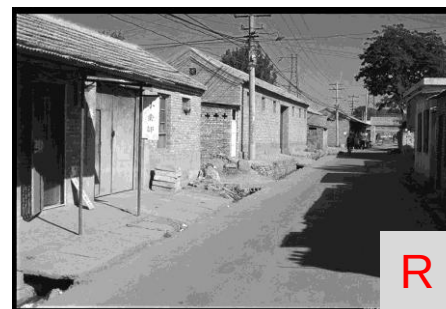


Image as array

- Color image
 - Three channels of R, G, B



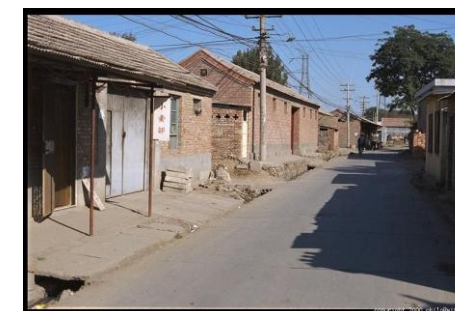
列 →

行 ↓

0.92	0.93	0.94	0.97	0.62	0.37	0.85	0.97	0.93	0.92	0.99
0.95	0.89	0.82	0.89	0.56	0.31	0.75	0.92	0.81	0.95	0.91
0.89	0.72	0.51	0.55	0.51	0.42	0.57	0.41	0.49	0.91	0.92
0.96	0.95	0.88	0.94	0.56	0.46	0.91	0.87	0.90	0.97	0.95
0.71	0.81	0.81	0.87	0.57	0.37	0.80	0.88	0.89	0.79	0.85
0.49	0.62	0.60	0.58	0.50	0.60	0.58	0.50	0.61	0.45	0.33
0.86	0.84	0.74	0.58	0.51	0.39	0.73	0.92	0.91	0.49	0.74
0.96	0.67	0.54	0.85	0.48	0.37	0.88	0.90	0.94	0.82	0.93
0.69	0.49	0.56	0.66	0.43	0.42	0.77	0.73	0.71	0.90	0.99
0.79	0.73	0.90	0.67	0.33	0.61	0.69	0.79	0.73	0.93	0.97
0.91	0.94	0.89	0.49	0.41	0.78	0.78	0.77	0.89	0.99	0.93

0.92	0.99
0.95	0.91
0.91	0.92
0.97	0.95
0.79	0.85
0.45	0.33
0.49	0.74
0.82	0.93
0.90	0.99
0.93	0.97
0.99	0.93

0.92	0.99
0.95	0.91
0.91	0.92
0.97	0.95
0.79	0.85
0.45	0.33
0.49	0.74
0.82	0.93
0.90	0.99
0.93	0.97
0.99	0.93

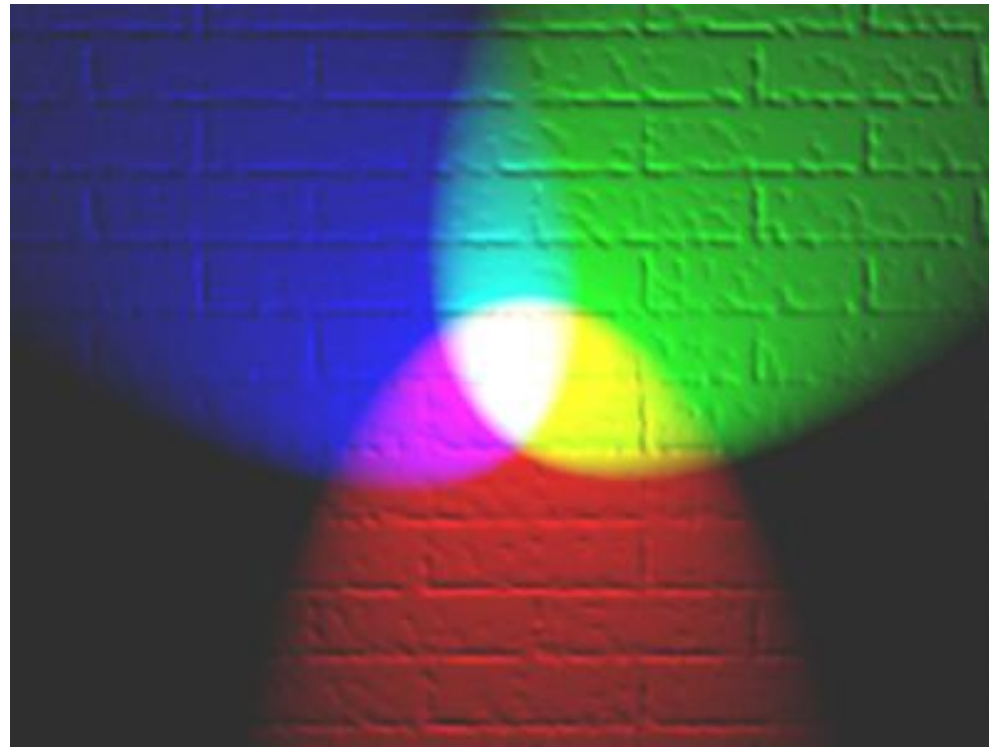


Color spaces

How can we represent color?

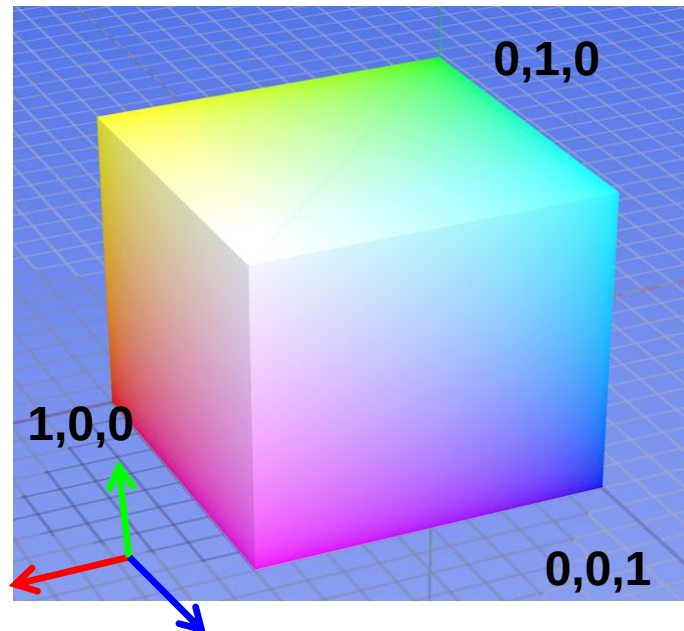
光学三原色 (RGB) : 红、绿、蓝 。

光学三原色混合后，组成显示屏显示颜色，三原色同时相加为白色



Color spaces: RGB

- Any color = $r \cdot R + g \cdot G + b \cdot B$
 - Strongly correlated channels
 - Non-perceptual



$R = 1$
($G=0, B=0$)



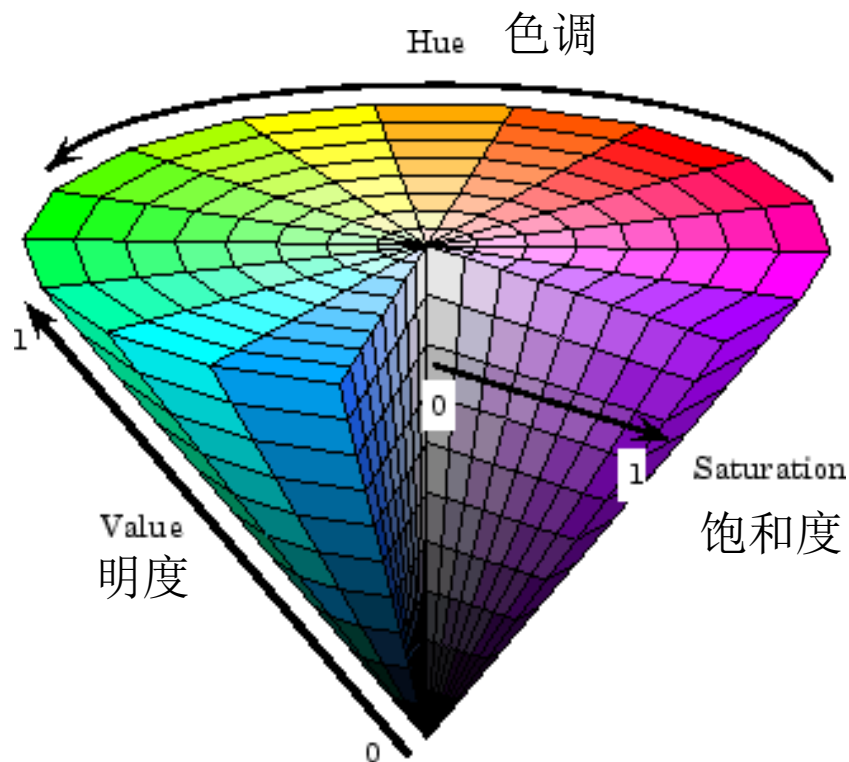
$G = 1$
($R=0, B=0$)



$B = 1$
($R=0, G=0$)

Color spaces: HSV

- Intuitive color space



更接近人对颜色的直观理解

色调 Hue: 取值0-360度

红色0度, 绿色120度, 蓝色240度

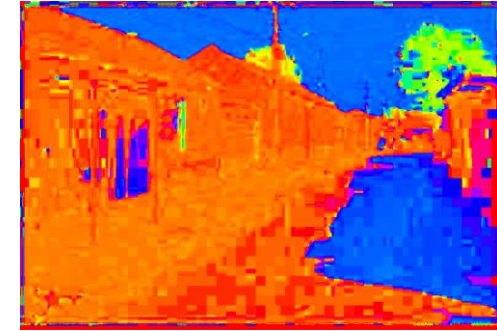
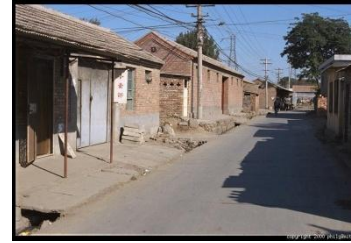
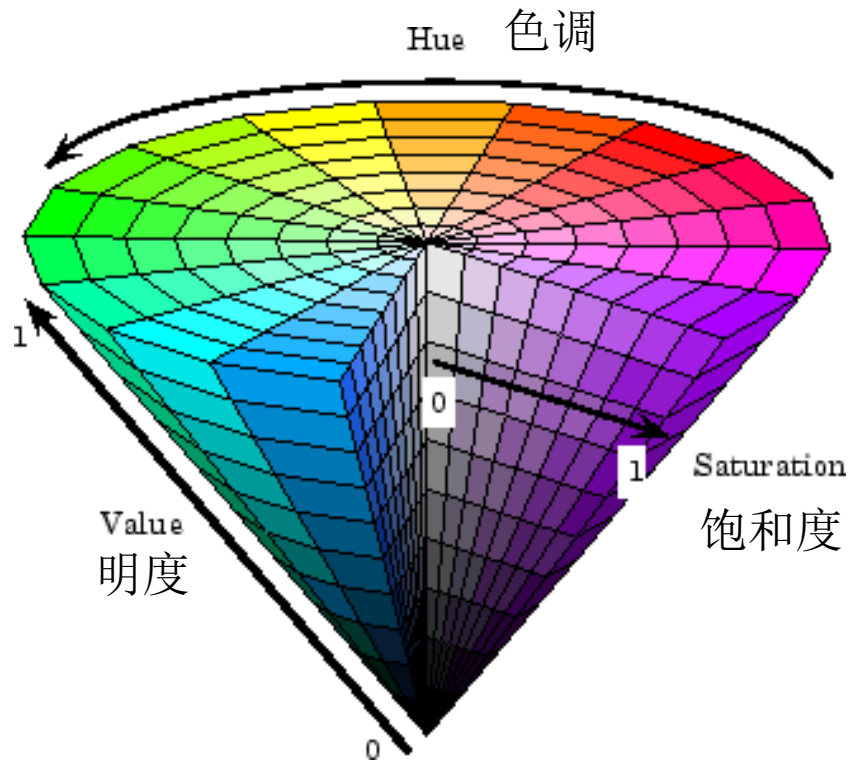
饱和度 Saturation: 颜色接近光谱色的程度, 取值0%-100%

取值为0代表黑白色, 无彩色信息

明度 Value: 颜色的明亮程度, 取值0%-100%

Color spaces: HSV

- Intuitive color space



H
(S=1,V=1)



S
(H=1,V=1)



V
(H=1,S=0)

Color spaces: YCbCr

- Good for digital television systems
 - Fast to compute

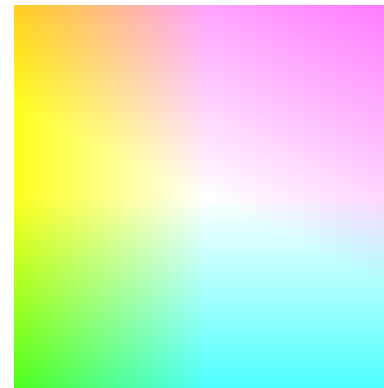
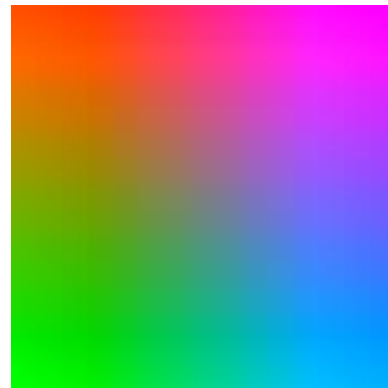
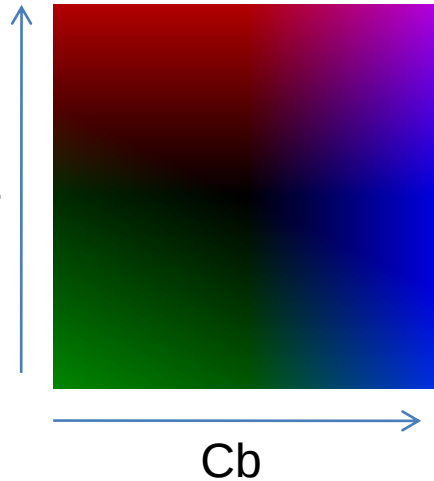


Y
(Cb=0.5,Cr=0.5)

Y=0

Y=0.5

Y=1



Cb
(Y=0.5,Cr=0.5)



Cr
(Y=0.5,Cb=0.5)

The Y component represents the intensity of the light. The Cb and Cr components indicate the intensities of the blue and red components relative to the green component.

目录



- 1, 相机模型
 - 相机内部参数、外部参数
 - 相机的组成
- 2, 图像与颜色
- 3, 图像处理基础

Image processing basics



- Point processing 点处理
 - Gamma correction
 - Histogram equalization
 - Window-center correction
- Filtering 滤波
 - Convolution, gradient
 - Mean, Gaussian, and median filters
 - Normalized cross-correlation (NCC)

Summary of image transformations



- An **image processing** operation (or transformation) typically defines a **new image g** in terms of an **existing image f** .

- Examples:

- Range transformation $g(x, y) = t(f(x, y))$

- Geometric transformation $g(x, y) = f(t_x(x, y), t_y(x, y))$

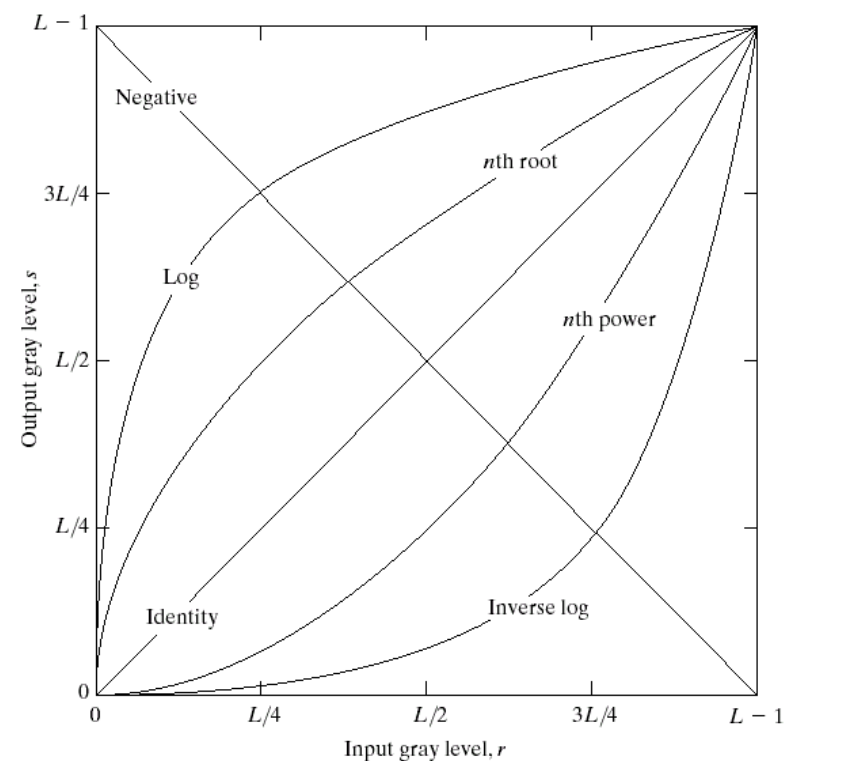
- Filtering (neighborhood processing) $g(x, y) = \int_{|u| < \varepsilon} \int_{|v| < \varepsilon} h(u, v) \cdot f(x - u, y - v) \cdot du \cdot dv$

Point processing

- For each image intensity value I , function $t()$ returns a transformed intensity

$$I' = t(I)$$

- Important:** every pixel is for itself
 - Spatial information is ignored!

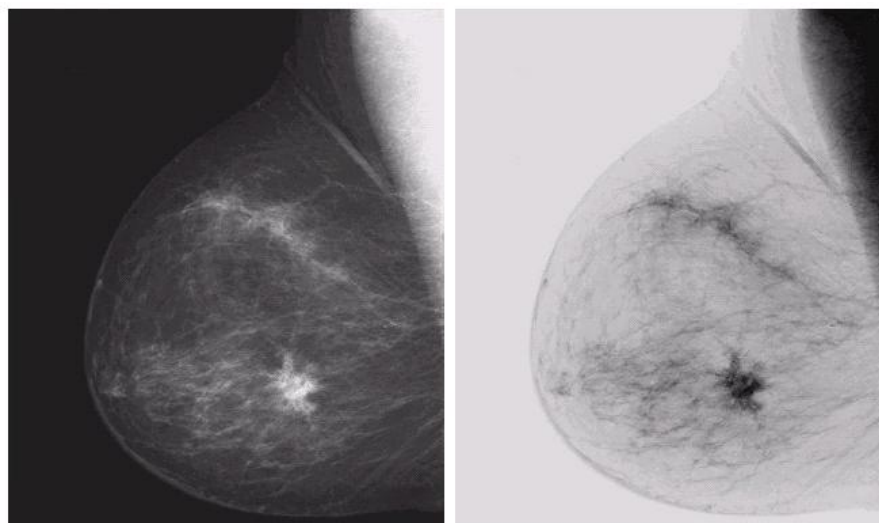


Point processing

- Negative

$$I' = 255 - I$$

负片



a b

FIGURE 3.4
(a) Original digital mammogram.
(b) Negative image obtained using the negative transformation in Eq. (3.2-1).
(Courtesy of G.E. Medical Systems.)

NOTE: we will often use notation I_p instead of $f(x,y)$ to denote intensity at pixel $p=(x,y)$

$$I_p = f(x, y)$$

$$I'_p = g(x, y)$$

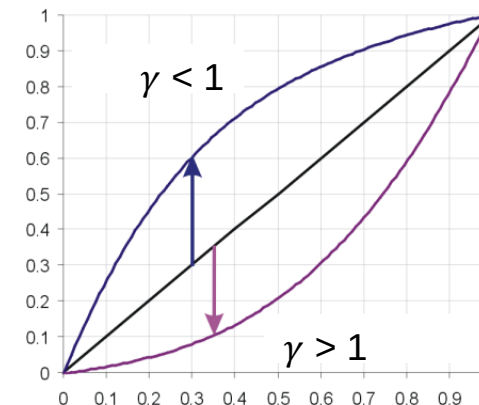
$$g(x, y) = t(f(x, y)) = 255 - f(x, y)$$

Point processing

- Gamma correction

伽马校正

$$y = f(x) = x^\gamma$$



原始图像



$\gamma = 0.5$



原始图像



$\gamma = 3.0$

Point processing

- Image histogram 直方图

Image Brightness

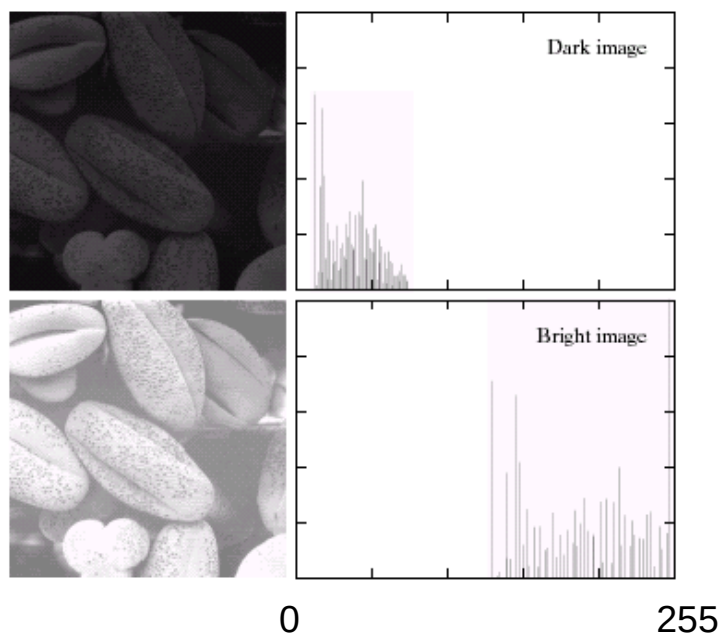
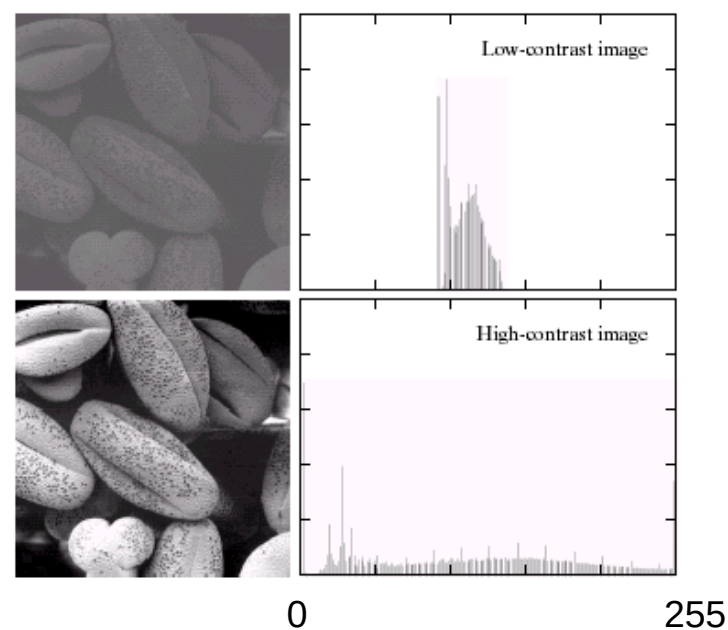


Image Contrast



probability of intensity i : $p_i = \frac{n_i}{n}$ ---number of pixels with intensity i
---total number of pixels in the image

小结



- 1, 相机模型
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 - 相机的组成
- 2, 图像与颜色
- 3, 图像处理基础
 - 伽马变换
 - 直方图

$$x = \phi_x u + \gamma v + \delta_x w$$

$$y = \phi_y v + \delta_y w$$

$$z = w$$

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = K \cdot [R | T] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

